Reinforcement Learning Algorithms for Solving Finite Horizon Mean Field Games

Derrick Chan and Elliott Hull
Mentor: JP Fouque
PHD Advisor: Mengrui Zhang

Mean Field Game

- An optimization problem involving one agent acting against a large number of identical agents (the mean field)
- We are searching for a Nash equilibrium
- A mean field game equilibrium is a set of control and distribution
    - Optimal move for each agent and distribution of the mean field

Mean Field Accumulation Problem:

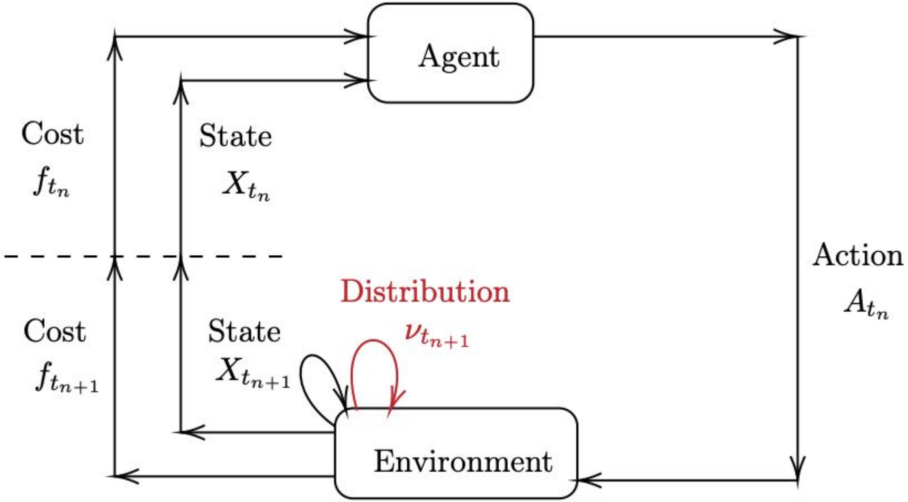
- The model consists of agents involved in a production activity where borrowing is <u>not allowed.</u>
- At time $t \in \mathbb{Z}^+$ the wealth of an agent is modeled by a process $X^{\alpha,\theta}$ with dynamics,

$$X_{t+1}^{\alpha,\theta} = G\left(\int a\, d\theta_t(a),\, W_t\right) \alpha_t$$

- $G\left(\int a\, d\theta_t(a), W_t\right)$ is the production function,
- $\theta = (\theta_t)_{0 \le t \le T}$ is the mean field term represented by the law of the investment level of the population,

Reinforcement Learning (RL):

the agent learns through trial and error through interactions with an environment.



Q-Learning:

a Q-matrix is updated to assign a value to each state-action pair based on the reward and discounted future rewards

Action Policy

- Choose the action that maximizes reward
- Explore vs Exploit dilemma
- Epsilon-greedy policy: $\pi(X_t) = \begin{cases} \operatorname{argmax}_a Q(X_t) & \text{with} \quad p = 1 - \epsilon \\ \text{Unif}\left[\mathcal{A}\right] & \text{with} \quad p = \epsilon \end{cases}$
- Soft-max policy: $\mathbb{P}(A_t = a | X_t = x) = \dfrac{\exp\{\phi * Q(x,a)\}}{\sum_{i \in \mathbb{A}} \exp\{\phi * Q(x,i)\}}$
- Adjusted soft-min policy

Outline of Code 1/3

N = 5 million episodes

Observe initial state $X^k_{t_0} \sim \mu_0$

T = 4 steps

Choose action $A^k_{t_n}$ by adjusted soft-min policy

Observe reward $f_{t_{n+1}} = f(X^k_{t_n}, A^k_{t_n}, \nu^k_{t_n})$

Observe new state $X^k_{t_{n+1}}$ from environment

Outline of Code 2/3

N = 5 million episodes, T = 4 steps

MFG: $\rho^{Q}_{n,x,a} >> \rho^{\mu}_{n}$

$\omega^{Q} = 0.55, \ \omega^{\mu} = 0.85$

$$\rho^{\nu}_{k} = \frac{1}{(1+k)^{\omega^{\nu}}}$$

$$\rho^{Q_n}_{x,a,k} = \frac{1}{(1 + N_T \#|(x,a,t_n,k)|)^{\omega^{Q}}}$$

Outline of Code 3/3

N = 5 million episodes, T = 4 steps

$$\nu^k_{t_n} = \nu^k_{t_n} + \rho^\nu_k (A^k_{t_n} - \nu^k_{t_n})$$

$$Q^k_n(x,a) :=$$

$$:= \begin{cases} Q^{k-1}_n(x,a) + \rho^{Q_n}_{x,a,k}[\mathcal{B} - Q^{k-1}_n(x,a)] & \text{if } (X^k_{t_n}, A^k_{t_n}) = (x,a) \\ Q^{k-1}_n(x,a) & \text{o.w.} \end{cases}$$

where

$$\mathcal{B} := \begin{cases} f_{t_{n+1}} + \gamma \min_{a' \in \mathcal{A}} Q^{k-1}_{n+1}(X^k_{t_{n+1}}, a'), & \text{if } t_n < T \\ f_{t_{n+1}}, & \text{o.w.} \end{cases}$$

Theoretical Solution

- From "A mean field capital accumulation game with HARA utility" (Huang, 2013)
- We have a fixed point equation that we computed in Python with FSolve to find optimal mean field investment
- Optimal control is linearly dependent on wealth

z_t : mean field investment at step t

c_t : optimal control coefficient at step t, $\hat{\alpha}_t(x) = c_t \cdot x_t$

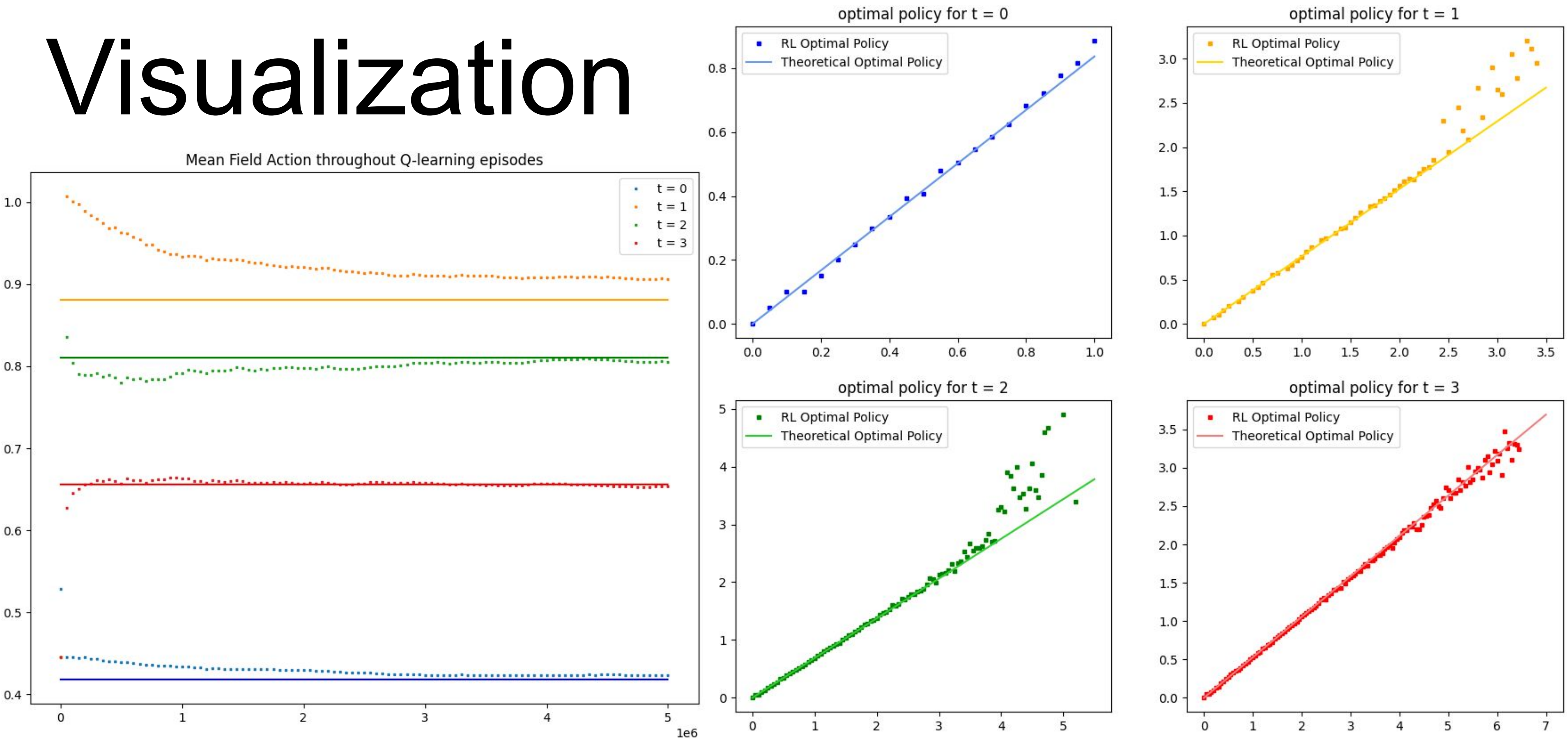
$$z_0 = 0.417897, \quad c_0 = 0.835794$$

$$z_1 = 0.880691, \quad c_1 = 0.763103$$

$$z_2 = 0.810105, \quad c_2 = 0.687734$$

$$z_3 = 0.655289, \quad c_3 = 0.527366$$

Visualization



Mean Field Control

- An optimization problem involving cooperative players
- In contrast to the Nash equilibrium we are finding the optimal behavior to **<u>minimize (maximize) a cost (reward).</u>**
- A central planner gives all identical agents an action policy
- Same problem different solution
    - I.e $\hat{\alpha} \neq \alpha^{\star}$ and $\hat{\nu} \neq \nu^{\star}$
- Harder to find solution

What’s Next

- Mean Field Control problems in infinite time horizon
- Mean Field Capital Accumulation Control problem in finite time horizon
- Further extension of Mean Field Game in finite time horizon
- With time, Mixed Mean Field control games

References

- Angiuli, Andrea, et al. “Reinforcement Learning for Mean Field Games, with Applications to Economics.” *ArXiv (Cornell University)*, 25 June 2021, https://doi.org/10.48550/arxiv.2106.13755. Accessed 1 Dec. 2025.
- Huang, Minyi. “A Mean Field Capital Accumulation Game with HARA Utility.” *Dynamic Games and Applications*, vol. 3, no. 4, 20 Aug. 2013, pp. 446–472, https://doi.org/10.1007/s13235-013-0092-9. Accessed 1 Dec. 2025.